

Optical Experiment Manuscript

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1 Introduction

In this manuscript, we will go through the detail of the optical experiment setup of the quantum circuit for Variational Quantum Eigensolver (VQE).

2 Components

Reference

Quarter-wave plate with fast axis at angle θ with respect to the horizontal axis (QWP):

$$e^{\frac{-i\pi}{4}} \begin{bmatrix} \cos^2(\theta) + i \sin^2(\theta) & (1-i) \sin(\theta) \cos(\theta) \\ (1-i) \sin(\theta) \cos(\theta) & \sin^2(\theta) + i \cos^2(\theta) \end{bmatrix} \quad (1)$$

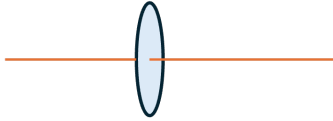


Figure 1: Quarter-wave plate

Quarter-wave plate with fast axis vertical (QWPv):

$$e^{\frac{i\pi}{4}} \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix} \quad (2)$$

Quarter-wave plate with fast axis horizontal (QWPh):

$$e^{\frac{-i\pi}{4}} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \quad (3)$$

Half-wave plate (HWP):

$$\begin{bmatrix} \cos(2\phi) & \sin(2\phi) \\ \sin(2\phi) & -\cos(2\phi) \end{bmatrix} \quad (4)$$

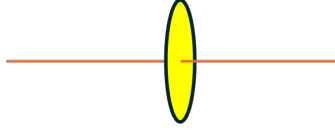


Figure 2: Half-wave plate

Dove Prism (DP) using Rotation matrix acting on the spatial mode:

$$\begin{bmatrix} \cos(\omega) & -\sin(\omega) \\ \sin(\omega) & \cos(\omega) \end{bmatrix} \quad (5)$$



Figure 3: Dove Prism

Beamsplitter (BS):

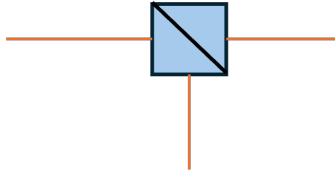


Figure 4: Beamsplitter

Polarizing beamsplitter (PBS):

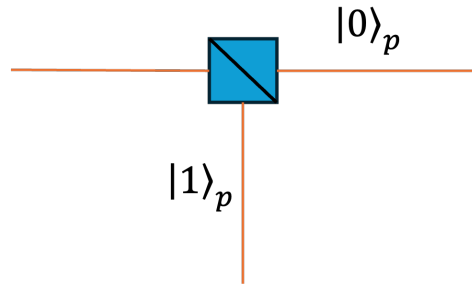


Figure 5: Polarizing Beamsplitter

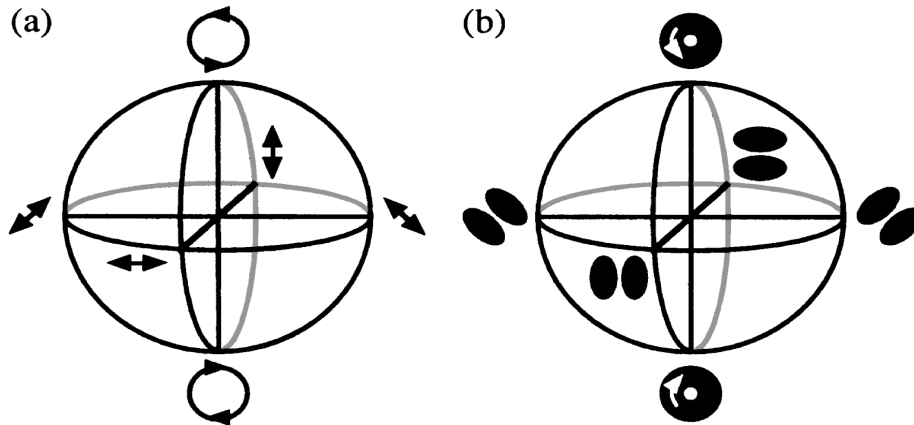


Figure 6: Poincare

For this experiment, we will use "01"/"10" Spatial Mode as our second qubit and Polarization

as our first qubit.

$|0\rangle = \text{Spatial Mode } 10 \text{ and Horizontal Polarization or } |0\rangle_s, |0\rangle_p$

$|1\rangle = \text{Spatial Mode } 01 \text{ and Vertical Polarization or } |1\rangle_s, |1\rangle_p$

$\frac{|0\rangle - i|1\rangle}{\sqrt{2}} = \text{South Pole of the Poincare Sphere or Left-circular Polarization } \frac{|L\rangle_s}{\sqrt{2}}, \frac{|L\rangle_p}{\sqrt{2}}$

$\frac{|0\rangle + i|1\rangle}{\sqrt{2}} = \text{North Pole of the Poincare Sphere or Right-circular Polarization } \frac{|R\rangle_s}{\sqrt{2}}, \frac{|R\rangle_p}{\sqrt{2}}$

$|+\rangle = \text{Diagonal Polarization } \frac{|D\rangle_s}{\sqrt{2}}, \frac{|D\rangle_p}{\sqrt{2}}$

$|-\rangle = \text{Anti-diagonal Polarization } \frac{|A\rangle_s}{\sqrt{2}}, \frac{|A\rangle_p}{\sqrt{2}}$

Therefore our initial state is $|0\rangle_p \otimes |0\rangle_s \equiv |0_p 0_s\rangle$

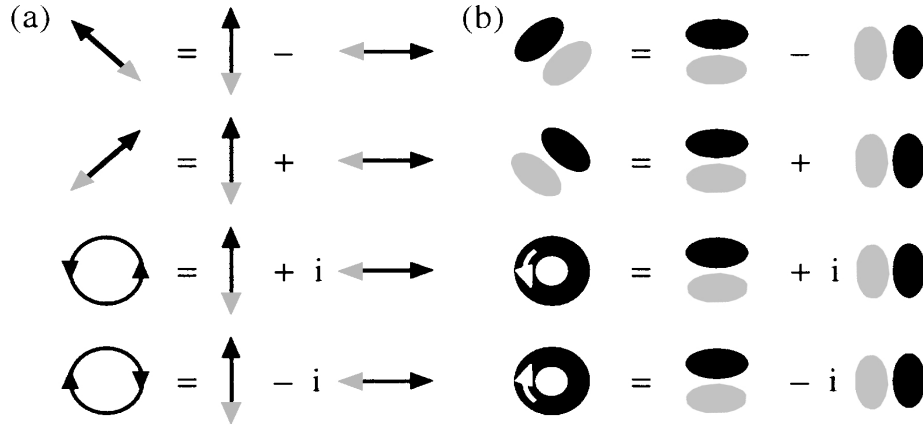


Figure 7: **Geometric Phase Shift**

3 Gates Realization

3.1 R_x Gate

$$R_x(\theta) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -i \sin(\frac{\theta}{2}) \\ -i \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix} \quad (6)$$

$$R_x\left(\frac{\pi}{2}\right) = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \quad (7)$$

$R_x(\frac{\pi}{2})$ acting on $|0\rangle$ gives:

$$R_x\left(\frac{\pi}{2}\right) |0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \frac{|0\rangle - i|1\rangle}{\sqrt{2}} \quad (8)$$

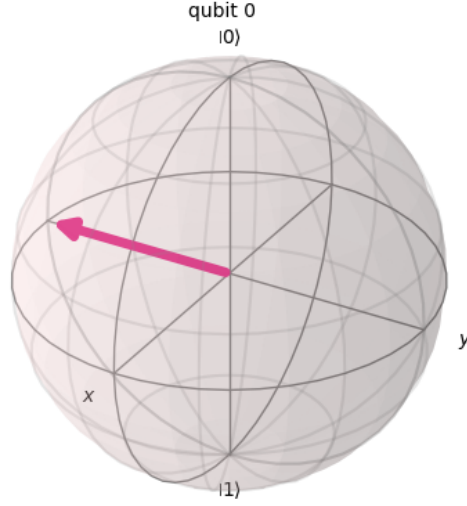


Figure 8: $\frac{|0\rangle - i|1\rangle}{\sqrt{2}}$

Quantum R_x gate can be realized for polarization by using quarter-wave plate with fast axis at angle θ with respect to the horizontal axis, where $\theta = \frac{\pi}{4}$:

$$\text{QWP}\left(\frac{\pi}{4}\right) = \begin{bmatrix} \cos^2(\frac{\pi}{4}) + i \sin^2(\frac{\pi}{4}) & (1-i) \sin(\frac{\pi}{4}) \cos(\frac{\pi}{4}) \\ (1-i) \sin(\frac{\pi}{4}) \cos(\frac{\pi}{4}) & \sin^2(\frac{\pi}{4}) + i \cos^2(\frac{\pi}{4}) \end{bmatrix} \quad (9)$$

$$= \begin{bmatrix} \frac{1}{2} + i\frac{1}{2} & (1-i)\frac{1}{2} \\ (1-i)\frac{1}{2} & \frac{1}{2} + i\frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1+i & (1-i) \\ (1-i) & 1+i \end{bmatrix} = \frac{1+i}{2} \begin{bmatrix} 1 & \frac{1-i}{1+i} \\ \frac{1-i}{1+i} & 1 \end{bmatrix} \quad (10)$$

$$= \frac{1+i}{2} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \quad (11)$$

We got the horizontal polarization $|0\rangle_p$ going through QWP at $\theta = \frac{\pi}{4}$ gives:

$$\text{QWP}\left(\frac{\pi}{4}\right)|0\rangle_p = \frac{1+i}{2} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \underbrace{\frac{1+i}{2} \begin{bmatrix} 1 \\ -i \end{bmatrix}}_{\text{Jones's vector}} = \frac{1+i}{2} |L\rangle_p \quad (12)$$

We can extract the time-dependent electric field from the Jones's vector:

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$$\vec{E}(t) = \text{Re} \{ |E\rangle e^{-i\omega t} \} \quad (13)$$

$$\Rightarrow \vec{E}_x(t) = \text{Re}(E_x e^{-i\omega t}) \quad (14)$$

$$\Rightarrow \vec{E}_y(t) = \text{Re}(E_y e^{-i\omega t}) \quad (15)$$

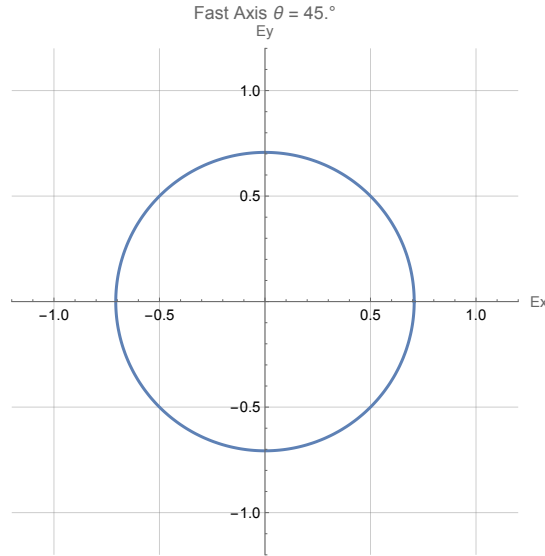


Figure 9: $\frac{|0\rangle - i|1\rangle}{\sqrt{2}}$ Time-dependent electric field

Now we need to figure out the realization of $R_x(\pi)$:

$$R_x(\pi) = \begin{bmatrix} \cos(\frac{\pi}{2}) & -i \sin(\frac{\pi}{2}) \\ -i \sin(\frac{\pi}{2}) & \cos(\frac{\pi}{2}) \end{bmatrix} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = -i \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ To realize } R_x(\pi), \text{ we}$$

3.2 Stokes vector measurement

The Stokes vector component can be calculated from E_x and E_y of the Jones vector:

$$\text{Stokes} = \begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix} = \begin{bmatrix} |E_x|^2 + |E_y|^2 \\ |E_x|^2 - |E_y|^2 \\ 2\text{Re}(E_x E_y^*) \\ 2\text{Im}(E_x E_y^*) \end{bmatrix} \quad (16)$$

Measuring the Stokes polarization parameters (Beth Schaefer)